

Speaker notes

Weekly report, 12 August 2026 | 16 slides

Slide 1 — Title

- Keep it to one sentence: two complete data cubes finished this week, and the headline is that anisotropy selects the d_{xy} pairing channel at half filling.

Slide 2 — Overview

Emergent altermagnetism and pairing

- Single band, spin-independent hopping, CPQMC at $T = 0$, benchmarked against exact diagonalisation.
- The one physical point worth making here: at $\delta = 0$ the two sublattices are related by a translation, so it is an ordinary antiferromagnet. At $\delta \neq 0$ only a 90 degree rotation relates them, which is the defining condition for altermagnetism. The form factor is then fixed by geometry to d_{xy} , because $\sin q_x \sin q_y$ is the C4-odd harmonic. That is symmetry, not a fit.

If asked: If asked why d_{xy} and not $d_{x^2-y^2}$: it is the only one of our four form factors that is odd under C4.

Slide 3 — Simulations

what ran and what it answered

- 300 node-hours across six nodes. 9,840 measurements, no gaps, no NaN.
- The sentence that matters is the fourth paragraph: pairing follows BOTH the magnetic order and the anisotropy, and the two cannot be separated. Everything after this slide is evidence for that claim.
- Twist-averaged boundary conditions are still running; results tomorrow.

If asked: If asked why it is not finished: a single-filling block only has 6 jobs, so most of each 32-core node sat idle. Fixed by splitting across anisotropies; the $U = 4$ half-filling check lands this morning.

Slide 4 — Method and observables

- Vertex = full minus bubble, integrated over imaginary time. Positive means the interaction is attractive in that channel.
- The equal-time vertex is CP-biased against ED, so we report the integrated form throughout.
- χ is an unnormalised double site sum and therefore extensive - every comparison across lattice sizes divides by N . Raw ratios matched N ratios to under 1%.
- The validation box is the strongest single argument that the machinery is right: at $U = 0$ the vertex vanishes to $1e-13$ in all four channels, and on-site s is repulsive in all 576 measured points.

If asked: If asked how you know the code is correct, point at the $U = 0$ null test before anything else.

Slide 5 — Anisotropy selects the d_{xy} channel

four channels x six fillings

- Read the columns. On-site s is the control - repulsive everywhere, magnitude shrinking 7x as n goes to 1, because U forbids double occupancy.
- Extended s and $d_{x^2-y^2}$ both need magnetism, so they only turn attractive above n about 0.73 and 0.56 respectively. Both are weakened by δ .
- d_{xy} is the one that reverses. Dip-then-recover at every filling, but the recovery only wins for $n \geq 0.889$: -0.013 to +0.021.
- One sentence to tie it: every channel needs magnetism to be attractive, and δ decides WHICH magnetic pattern you get - square-lattice antiferromagnetic, which feeds extended s and $d_{x^2-y^2}$, or altermagnetic d_{xy} .

If asked: Caveat if pressed: the $n = 0.5$ row is the least reliable - largest values, and $L = 8$ separates from the other two. Restrict claims to $n \geq 0.667$.

Slide 6 — Pairing vertex vs U

$n = 1$ against $n = 0.778$

- Left, half filling: at $\delta = 0$ and 0.1 the red d_{xy} curve is below zero, the worst of the four. At $\delta = 0.2$ it crosses; at 0.3 it passes $d_{x^2-y^2}$; from 0.4 it is the top curve.
- Right, doped: d_{xy} NEVER crosses zero. The same anisotropy drives it further into repulsion, -0.012 down to -0.042 at $\delta = 0.4$.

- The contrast is the argument. Showing only half filling invites the question 'how do you know delta is not just helping d_{xy} everywhere?' The doped panel answers it before it is asked.

If asked: Both figures: every curve starts at exactly zero at $U = 0$, and all peak near $U = 4-6$, not at the strongest coupling.

Slide 7 — Pairing vertex vs n

$U = 4$, one panel per anisotropy

- The cleanest new fact here is that the four crossings converge. At $\delta = 0$ they are spread from $n = 0.53$ to 0.73 ; by $\delta = 0.7$ they are bunched between 0.75 and 0.82 . Anisotropy homogenises the channels and pushes every attractive window toward half filling.
- The d_{xy} inversion is worth pointing at directly: in the $\delta = 0$ panel red is the HIGHEST curve at $n = 0.5$; by $\delta = 0.7$ it is the LOWEST at that same filling. Same channel, same filling, opposite sign, only delta changed.

If asked: The fanning-out is the mechanism showing itself: where the system is magnetic, delta redistributes weight between channels; where it is not, delta is pure frustration and drags everything down together.

Slide 8 — Vertex vs U , all anisotropies

$n = 1$

- This is the same data as slide 6 left, expanded. Point at the red curve rising through the panels: below zero at $\delta = 0$ and 0.1 , crossing at 0.2 , top curve from 0.4 .
- The whole reordering happens over a narrow window in anisotropy, between 0.1 and 0.4 .

If asked: If asked why the peak is at $U = 4-6$ rather than $U = 8$: attraction is mediated by magnetic correlations, which grow with U , but strong U also localises the carriers. The product peaks in between.

Slide 9 — Why extended s leads

the one place we depart from White

- Be upfront: White finds $d_{x^2-y^2}$ the most attractive channel. We reproduce all four of his channel SIGNS, but our ordering differs - extended s is larger, 0.0364 against 0.0277 at $U = 4$.
- Huang, Lin and Gubernatis explain it: extended s dominates the LOCAL pairing while $d_{x^2-y^2}$ dominates the LONG-RANGE part. Our χ integrates over all separations, so the local piece wins on magnitude.
- Separating them needs the real-space $P(R)$ decomposition, which we have not built yet. That is the next piece of work.

If asked: This is a disagreement with a published explanation, not an unexplained discrepancy. Say it that way.

Slide 10 — The effect switches off below $n \sim 0.85$

- d_{xy} at $L = 12$, $U = 4$: reversed at $n = 0.5$ and 0.556 , no swap at 0.667 and 0.778 , and the SWAP appears at 0.889 and 1.000 .
- The second table shows anisotropy pushing every channel's attractive window toward half filling - $d_{x^2-y^2}$ turns attractive from $n = 0.53$ at $\delta = 0$ but only from 0.75 at $\delta = 0.7$.
- The pairing boundary sits between $n = 0.778$ and 0.889 , the same place the magnetic moment becomes strong. The two hang together.

If asked: Scoping the claim to $n \geq 0.85$ is a strength, not a weakness - it is falsifiable, with a control region where the effect vanishes.

Slide 11 — Phase diagram

every channel, every interaction, one shared colour scale

- Start with the top row: the whole $U = 0$ row is one flat colour. That is the null test seen directly.
- Then the columns. On-site s uniformly dark - zero attractive area at every U . Extended s has a vertical boundary near $n = 0.7$ that barely moves. $d_{x^2-y^2}$ is almost fully attractive at $U = 2$ and retreats as U grows. d_{xy} is the only column with a corner rather than a stripe.
- The number that carries it: attractive area goes from 100% to 42% for $d_{x^2-y^2}$ as U goes 2 to 8 , while d_{xy} goes 22% to 35% . As U grows, $d_{x^2-y^2}$ loses territory and d_{xy} gains it.

If asked: Colour scale is symmetric-log because on-site s reaches -55 while $d_{x^2-y^2}$ spans only -11 to $+6$. Equal colour steps are NOT equal differences in χ - use it to show where signs change, not magnitudes.

Slide 12 — Dominating channel

- Which channel has the largest vertex at each point. On-site s never wins anywhere - it is the most repulsive channel at every single point.
 - $d_{x^2-y^2}$ owns the doped interior; d_{xy} wins in the corners - far doping at low δ , and half filling at high δ .
- If asked: If cells are marked as unresolved: those are points where the leading channel is within 2 sigma of the runner-up, so the colour there is a coin flip. Most sit in the low-filling corner.*

Slide 13 — All four channels on one scale

P / P_{bub}

- Dividing by the bubble removes the kinematic factor that made the y-axes differ by 6x, so all four fit on one plot and the question becomes: above or below 1.
- All four curves start at exactly 1.000 at $U = 0$.
- At $U = 8$ the interaction enhances $d_{x^2-y^2}$ by 26% and extended s by 21%, while suppressing d_{xy} by 8% and on-site s by 46%. Extended s leads up to U about 6, then $d_{x^2-y^2}$ overtakes.

If asked: This is $\delta = 0$ only - the limit where the square-lattice benchmarks apply.

Slide 14 — AM order parameter against U and δ

meeting point 7

- The two contributions pull in opposite directions. U builds the moment, roughly linearly: 0.0055 at $U = 2$ to 0.0257 at $U = 8$. δ destroys it: the same quantity falls by a factor of three as δ goes 0 to 0.7.
- The product plateaus past δ about 0.3 - that is two effects cancelling, not saturation of one.

If asked: Be honest about two limits: $U = 6$ and $U = 8$ are within 1 sigma of each other at $\delta \geq 0.6$, so the U dependence is established only up to δ about 0.4. And Δ_m is still shrinking with lattice size at $L = 12$, so these are finite-size magnitudes.

Slide 15 — A correction we found ourselves

the magnetic peak leaves (π, π)

- Stress that we found this by checking our own order parameter, not because a referee asked - and the conclusion survived.
- The peak is incommensurate at small δ , commensurate Neel near $\delta = 0.35$, and stripe-like at large δ . At $\delta = 0.7$, fixed (π, π) captures only 80% of the magnetic weight.
- Recomputing at the true peak leaves the suppression intact - factor 2.3 instead of 3.0. The headline survives.
- Both limits match published work: incommensurate structures in Huang et al., stripe phases in the t - t' - δ paper.

If asked: This slide buys credibility for everything else. A group that finds its own errors is trusted on the rest.

Slide 16 — Literature check

the field has moved

- Two 2025-26 papers already claim altermagnetism on this lattice family. 'Checkerboard hosts altermagnetism' is no longer available as a headline - arXiv 2607.06106 is five weeks old and says exactly that.
- What it gives us: the t - t' - δ paper explicitly declines to compute pairing. Our d_{xy} channel swap is unclaimed, it is CPQMC at intermediate coupling where the competitors are mean field, and we have the doping axis nobody else has.
- Recommendation: reframe to lead with pairing, use magnetism as the explanation. That is a stronger position than the original plan, but it means rewriting the introduction.

If asked: This is a decision for Prof Tang, not a conclusion. Put it to him as a question.

Numbers worth having memorised

- 9,840 measurements, 300 node-hours, six nodes.
- $U = 0$ null test: vertex vanishes to $1e-13$ in all four channels.
- d_{xy} channel swap: -0.010 to +0.026 at $U = 4$, half filling; 27 sigma; identical at $L = 8, 10, 12$.
- Filling boundary between $n = 0.778$ and 0.889 .
- Attractive area as U goes 2 to 8: $d_{x^2-y^2}$ 100% to 42%, d_{xy} 22% to 35%.

Three things to volunteer before being asked

- χ/N is flat in system size, so there is NO tendency to long-range order. What we can defend is a change in which channel is attractive, not superconductivity. Huang et al. reach the same conclusion.
- $d_{x^2-y^2}$ is the noisiest channel in the dataset: 20% seed noise against 0.3-5% elsewhere.
- The $n = 0.5$ corner has the largest values and the worst size collapse - restrict claims to $n \geq 0.667$.